

Fernando Garcia

Staff Software Architect • Technical Leader • Software Engineer

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Staff Software Architect focused on legacy modernization, clean architecture, distributed systems, and practical engineering leadership.

Highlights

- ⚡ **15+ Years** building software across FinTech, GovTech, telecom, insurance, gaming, and enterprise platforms
- ⚡ **Legacy Modernization:** monoliths to modular, testable, event-driven architectures
- ⚡ **Event-Driven Systems:** AWS SNS/SQS, Lambda, Kafka, PostgreSQL, and high-concurrency backends
- ⚡ **Engineering Leadership:** clean architecture, TDD, CI/CD, DevSecOps, and team mentoring

EXECUTIVE SUMMARY

More than 15 years of experience across software engineering, technical leadership, and staff-level architecture roles, building and modernizing software in high-concurrency environments. Focused on migrating legacy systems into modular, event-driven architectures based on Clean Architecture using Java, Kotlin, Python, AWS, Kubernetes, and DevSecOps practices. Experienced in leading teams, improving core business systems, and designing solutions around databases, BI, Data Warehouse, and advanced analytics. Comfortable turning complex technical problems into maintainable platforms that teams can evolve with confidence.

CORE EXPERTISE

- Engineering Platforms & Productivity
- Distributed Systems & Event-Driven Architecture
- Legacy Modernization & Clean Architecture
- Databases, BI & Data Warehouse
- Application Security & DevSecOps
- Engineering Standards, Technical Documentation & Architecture Decision Records (ADRs)

SELECTED ACHIEVEMENTS

- Legacy-to-Platform Modernization:** Led migrations from monolithic systems to modular, cloud-native services across FinTech and GovTech environments.
- Core System Improvements:** Designed event-driven solutions using AWS SNS/SQS, Lambda, Kafka, PostgreSQL, and Java/Kotlin services to improve resilience, traceability, and throughput.
- Engineering Standards:** Introduced clean architecture, hexagonal architecture, TDD, CI/CD, DevSecOps gates, and technical decision records to make delivery more predictable.
- Practical AI Adoption:** Helped scale secure AI-assisted engineering workflows at Mercado Libre from 0% to 30% adoption using reusable templates and automation.

PROFESSIONAL EXPERIENCE

Mercado Libre | Technical Leader

June 2025 – Present

- **Engineering Standards:** Led security and innovation initiatives across large-scale APIs and microservices, defining architecture practices that improved maintainability and delivery quality.
- **Developer Productivity:** Designed reusable engineering templates and automation workflows that helped teams adopt secure, consistent development practices.
- **Practical AI Enablement:** Built internal AI-assisted workflows using Model Context Protocol (MCP), Claude Skills, and n8n to support operational bots and developer tooling.
- **Architecture Governance:** Promoted clean/hexagonal architecture, TDD, SDD, technical documentation, and ADR-based decision-making to keep services easier to test, review, and evolve.

Streambe | External Software Consultant

August 2016 – Present

- **Distributed Architecture:** Designed and implemented decoupled, event-driven architectures leveraging AWS SNS, SQS, and Lambda, optimizing monolithic core processing.
- **Mentorship & Standards:** Provided technical leadership and mentorship to engineering teams, spearheading architectural design patterns and introducing robust CI/CD practices via GitLab-CI.

Workia | Staff Engineer & Software Architect

August 2023 – June 2025

- **Data Engineering Platform:** Led the design and implementation of a scalable Data Warehouse from scratch using event-driven loading pipelines (Java 17, PostgreSQL, AWS Redshift, Pentaho), drastically reducing computing resource consumption.
- **Payroll Core Optimization:** Engineered a high-concurrency, in-memory payroll processing and auditing system under clean architecture principles, introducing an auditable dry-run liquidation engine.

Santander Tecnología Argentina | Senior Software Engineer

July 2022 – August 2023

- **FinTech Cloud Modernization:** Developed core investment platform modules for Digital Broker, with a specific focus on Mutual Fund operations using Kotlin (Ktor) and Node.js deployed over Kubernetes (OpenShift).
- **Operational Resilience:** Contributed to stabilizing Mutual Fund operations, improving resilience, maintainability, API reliability, and support for large data volumes.

AFIP | Software Engineer & DevSecOps Practitioner

January 2020 – August 2022

- **DevSecOps Automation:** Spearheaded the end-to-end automation of a private cloud (bare-metal) CI/CD pipeline incorporating security gates (Defect Dojo, Trivy, SonarQube), ArgoCD, Jenkins, and Kubernetes.
- **Enterprise Refactoring:** Led technological migrations applying 12-Factor App methodology to legacy transactional architectures (Java 17, Spring Boot, Vue.js, Docker).

Previous Professional Experience (Summary)

- **Boehringer Ingelheim** (*Regional Data Science Specialist, 2019*): Conducted advanced exploratory analysis and predictive statistics using RStudio and Python.
- **Etermax** (*Software Engineer & Big Data Developer, 2016 - 2019*): Built predictive Lifetime Value (LTV) data models, high-throughput Kotlin backends, and C#/Unity product prototypes.
- **Universidad Nacional de San Martín** (*BI Leader, 2012 - 2016*): Managed the institution's BI and Data Warehouse ecosystem, represented the university in CONICET user committees, and improved SIU open-source applications.
- **Early Career (2004 - 2012)**: Extensive engineering background across **Mapfre**, **Telefonica**, **Laboratorios Raffo**, and public science, technology, and education institutions, specializing in Java development, database modeling

(Oracle, Postgres), Business Intelligence, and technical support.

ENGINEERING PHILOSOPHY

- **Design Standards:** DDD (Domain-Driven Design), Hexagonal/Clean Architecture, SOLID, 12-Factor Apps.
- **Developer Experience (DX):** Automated Security Gates, CI/CD Pipelines, Technical Documentation, Architecture Decision Records (ADRs).
- **Quality Assurance:** TDD (Test-Driven Development), BDD, Automated Refactoring, Structured Peer Reviews.

TECHNICAL STACK

- **Languages & Core:** Java (8/17), Kotlin (Ktor), Python, Go, Scala, JavaScript/TypeScript, SQL.
- **Cloud & Infrastructure:** AWS (Lambda, ECS, SQS, Redshift), Kubernetes, OpenShift, Docker, Terraform.
- **AI & Automation:** Model Context Protocol (MCP), Claude Skills, n8n, RStudio.
- **CI/CD & Observability:** GitLab-CI, Jenkins, ArgoCD, Prometheus, Grafana, CloudWatch.

EDUCATION & CERTIFICATIONS

- **Master's Degree in Data Mining & Knowledge Discovery** – Universidad de Buenos Aires (2008 – 2010)
- **Information Systems Engineer (Ingeniero en Informática)** – Universidad Católica de Salta (2003 – 2005)
- **Mastering Microservices with Java** – Professional Certification.